

Operators on Inner Product Spaces

The deepest results related to inner product spaces deal with the subject to which we now turn—linear maps and operators on inner product spaces. As we will see, good theorems can be proved by exploiting properties of the adjoint.

The hugely important spectral theorem will provide a complete description of self-adjoint operators on real inner product spaces and of normal operators on complex inner product spaces. We will then use the spectral theorem to help understand positive operators and unitary operators, which will lead to unitary matrices and matrix factorizations. The spectral theorem will also lead to the popular singular value decomposition, which will lead to the polar decomposition.

The most important results in the rest of this book are valid only in finite dimensions. Thus from now on we assume that V and W are finite-dimensional.

standing assumptions for this chapter

- F denotes $\mathbf{R}$ or $\mathbf{C}$.
- V and W are nonzero finite-dimensional inner product spaces over F .



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Market square in Lviv, a city that has had several names and has been in several countries because of changing international borders. From 1772 until 1918, the city was in Austria and was called Lemberg. Between World War I and World War II, the city was in Poland and was called Lwów. During this time, mathematicians in Lwów, particularly Stefan Banach (1892–1945) and his colleagues, developed the basic results of modern functional analysis, using tools of analysis to study infinite-dimensional vector spaces.

Since the end of World War II, Lviv has been in Ukraine, which was part of the Soviet Union until Ukraine became an independent country in 1991.

7A Self-Adjoint and Normal Operators

Adjoint

7.1 definition: *adjoint*, T^*

Suppose $T \in \mathcal{L}(V, W)$. The *adjoint* of T is the function $T^*: W \rightarrow V$ such that

$$\langle Tv, w \rangle = \langle v, T^*w \rangle$$

for every $v \in V$ and every $w \in W$.

To see why the definition above makes sense, suppose $T \in \mathcal{L}(V, W)$. Fix $w \in W$. Consider the linear functional

$$v \mapsto \langle Tv, w \rangle$$

on V that maps $v \in V$ to $\langle Tv, w \rangle$; this

linear functional depends on T and w . By the Riesz representation theorem (6.42), there exists a unique vector in V such that this linear functional is given by taking the inner product with it. We call this unique vector T^*w . In other words, T^*w is the unique vector in V such that

$$\langle Tv, w \rangle = \langle v, T^*w \rangle$$

for every $v \in V$.

In the equation above, the inner product on the left takes place in W and the inner product on the right takes place in V . However, we use the same notation $\langle \cdot, \cdot \rangle$ for both inner products.

*The word **adjoint** has another meaning in linear algebra. In case you encounter the second meaning elsewhere, be warned that the two meanings for adjoint are unrelated to each other.*

7.2 example: *adjoint of a linear map from $\mathbf{R}^3$ to $\mathbf{R}^2$*

Define $T: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ by

$$T(x_1, x_2, x_3) = (x_2 + 3x_3, 2x_1).$$

To compute T^* , suppose $(x_1, x_2, x_3) \in \mathbf{R}^3$ and $(y_1, y_2) \in \mathbf{R}^2$. Then

$$\begin{aligned} \langle T(x_1, x_2, x_3), (y_1, y_2) \rangle &= \langle (x_2 + 3x_3, 2x_1), (y_1, y_2) \rangle \\ &= x_2y_1 + 3x_3y_1 + 2x_1y_2 \\ &= \langle (x_1, x_2, x_3), (2y_2, y_1, 3y_1) \rangle. \end{aligned}$$

The equation above and the definition of the adjoint imply that

$$T^*(y_1, y_2) = (2y_2, y_1, 3y_1).$$

7.3 example: *adjoint of a linear map with range of dimension at most 1*

Fix $u \in V$ and $x \in W$. Define $T \in \mathcal{L}(V, W)$ by

$$Tv = \langle v, u \rangle x$$

for each $v \in V$. To compute T^* , suppose $v \in V$ and $w \in W$. Then

$$\begin{aligned} \langle Tv, w \rangle &= \langle \langle v, u \rangle x, w \rangle \\ &= \langle v, u \rangle \langle x, w \rangle \\ &= \langle v, \langle w, x \rangle u \rangle. \end{aligned}$$

Thus

$$T^*w = \langle w, x \rangle u.$$

In the two examples above, T^* turned out to be not just a function from W to V but a linear map from W to V . This behavior is true in general, as shown by the next result.

The two examples above and the proof below use a common technique for computing T^ : start with a formula for $\langle Tv, w \rangle$ then manipulate it to get just v in the first slot; the entry in the second slot will then be T^*w .*

7.4 *adjoint of a linear map is a linear map*

If $T \in \mathcal{L}(V, W)$, then $T^* \in \mathcal{L}(W, V)$.

Proof Suppose $T \in \mathcal{L}(V, W)$. If $v \in V$ and $w_1, w_2 \in W$, then

$$\begin{aligned} \langle Tv, w_1 + w_2 \rangle &= \langle Tv, w_1 \rangle + \langle Tv, w_2 \rangle \\ &= \langle v, T^*w_1 \rangle + \langle v, T^*w_2 \rangle \\ &= \langle v, T^*w_1 + T^*w_2 \rangle. \end{aligned}$$

The equation above shows that

$$T^*(w_1 + w_2) = T^*w_1 + T^*w_2.$$

If $v \in V$, $\lambda \in \mathbf{F}$, and $w \in W$, then

$$\begin{aligned} \langle Tv, \lambda w \rangle &= \overline{\lambda} \langle Tv, w \rangle \\ &= \overline{\lambda} \langle v, T^*w \rangle \\ &= \langle v, \lambda T^*w \rangle. \end{aligned}$$

The equation above shows that

$$T^*(\lambda w) = \lambda T^*w.$$

Thus T^* is a linear map, as desired. ■

7.5 properties of the adjoint

Suppose $T \in \mathcal{L}(V, W)$. Then

- (a) $(S + T)^* = S^* + T^*$ for all $S \in \mathcal{L}(V, W)$;
- (b) $(\lambda T)^* = \bar{\lambda}T^*$ for all $\lambda \in \mathbf{F}$;
- (c) $(T^*)^* = T$;
- (d) $(ST)^* = T^*S^*$ for all $S \in \mathcal{L}(W, U)$ (here U is a finite-dimensional inner product space over $\mathbf{F}$);
- (e) $I^* = I$, where I is the identity operator on V ;
- (f) if T is invertible, then T^* is invertible and $(T^*)^{-1} = (T^{-1})^*$.

Proof Suppose $v \in V$ and $w \in W$.

- (a) If $S \in \mathcal{L}(V, W)$, then

$$\begin{aligned}\langle (S + T)v, w \rangle &= \langle Sv, w \rangle + \langle Tv, w \rangle \\ &= \langle v, S^*w \rangle + \langle v, T^*w \rangle \\ &= \langle v, S^*w + T^*w \rangle.\end{aligned}$$

Thus $(S + T)^*w = S^*w + T^*w$, as desired.

- (b) If $\lambda \in \mathbf{F}$, then

$$\langle (\lambda T)v, w \rangle = \lambda \langle Tv, w \rangle = \lambda \langle v, T^*w \rangle = \langle v, \bar{\lambda}T^*w \rangle.$$

Thus $(\lambda T)^*w = \bar{\lambda}T^*w$, as desired.

- (c) We have

$$\langle T^*w, v \rangle = \overline{\langle v, T^*w \rangle} = \overline{\langle Tv, w \rangle} = \langle w, Tv \rangle.$$

Thus $(T^*)^*v = Tv$, as desired.

- (d) Suppose $S \in \mathcal{L}(W, U)$ and $u \in U$. Then

$$\langle (ST)v, u \rangle = \langle S(Tv), u \rangle = \langle Tv, S^*u \rangle = \langle v, T^*(S^*u) \rangle.$$

Thus $(ST)^*u = T^*(S^*u)$, as desired.

- (e) Suppose $u \in V$. Then

$$\langle Iu, v \rangle = \langle u, v \rangle.$$

Thus $I^*v = v$, as desired.

- (f) Suppose T is invertible. Take adjoints of both sides of the equation $T^{-1}T = I$, then use (d) and (e) to show that $T^*(T^{-1})^* = I$. Similarly, the equation $TT^{-1} = I$ implies $(T^{-1})^*T^* = I$. Thus $(T^{-1})^*$ is the inverse of T^* , as desired. ■

If $\mathbf{F} = \mathbf{R}$, then the map $T \mapsto T^*$ is a linear map from $\mathcal{L}(V, W)$ to $\mathcal{L}(W, V)$, as follows from (a) and (b) of the result above. However, if $\mathbf{F} = \mathbf{C}$, then this map is not linear because of the complex conjugate that appears in (b).

The next result shows the relationship between the null space and the range of a linear map and its adjoint.

7.6 null space and range of T^*

Suppose $T \in \mathcal{L}(V, W)$. Then

- (a) $\text{null } T^* = (\text{range } T)^\perp$;
- (b) $\text{range } T^* = (\text{null } T)^\perp$;
- (c) $\text{null } T = (\text{range } T^*)^\perp$;
- (d) $\text{range } T = (\text{null } T^*)^\perp$.

Proof We begin by proving (a). Let $w \in W$. Then

$$\begin{aligned} w \in \text{null } T^* &\iff T^*w = 0 \\ &\iff \langle v, T^*w \rangle = 0 \text{ for all } v \in V \\ &\iff \langle Tv, w \rangle = 0 \text{ for all } v \in V \\ &\iff w \in (\text{range } T)^\perp. \end{aligned}$$

Thus $\text{null } T^* = (\text{range } T)^\perp$, proving (a).

If we take the orthogonal complement of both sides of (a), we get (d), where we have used 6.52. Replacing T with T^* in (a) gives (c), where we have used 7.5(c). Finally, replacing T with T^* in (d) gives (b). ■

As we will soon see, the next definition is intimately connected to the matrix of the adjoint of a linear map.

7.7 definition: conjugate transpose, A^*

The *conjugate transpose* of an m -by- n matrix A is the n -by- m matrix A^* obtained by interchanging the rows and columns and then taking the complex conjugate of each entry. In other words, if $j \in \{1, \dots, n\}$ and $k \in \{1, \dots, m\}$, then

$$(A^*)_{j,k} = \overline{A_{k,j}}.$$

7.8 example: conjugate transpose of a 2-by-3 matrix

The conjugate transpose of the 2-by-3 matrix $\begin{pmatrix} 2 & 3+4i & 7 \\ 6 & 5 & 8i \end{pmatrix}$ is the 3-by-2 matrix

$$\begin{pmatrix} 2 & 6 \\ 3-4i & 5 \\ 7 & -8i \end{pmatrix}.$$

If a matrix A has only real entries, then $A^* = A^t$, where A^t denotes the transpose of A (the matrix obtained by interchanging the rows and the columns).

The next result shows how to compute the matrix of T^* from the matrix of T .
Caution: With respect to nonorthonormal bases, the matrix of T^* does not necessarily equal the conjugate transpose of the matrix of T .

The adjoint of a linear map does not depend on a choice of basis. Thus we frequently emphasize adjoints of linear maps instead of transposes or conjugate transposes of matrices.

7.9 matrix of T^* equals conjugate transpose of matrix of T

Let $T \in \mathcal{L}(V, W)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $f_1, \dots, f_m$ is an orthonormal basis of W . Then $\mathcal{M}(T^*, (f_1, \dots, f_m), (e_1, \dots, e_n))$ is the conjugate transpose of $\mathcal{M}(T, (e_1, \dots, e_n), (f_1, \dots, f_m))$. In other words,

$$\mathcal{M}(T^*) = (\mathcal{M}(T))^*.$$

Proof In this proof, we will write $\mathcal{M}(T)$ and $\mathcal{M}(T^*)$ instead of the longer expressions $\mathcal{M}(T, (e_1, \dots, e_n), (f_1, \dots, f_m))$ and $\mathcal{M}(T^*, (f_1, \dots, f_m), (e_1, \dots, e_n))$.

Recall that we obtain the k^{th} column of $\mathcal{M}(T)$ by writing Te_k as a linear combination of the f_j 's; the scalars used in this linear combination then become the k^{th} column of $\mathcal{M}(T)$. Because $f_1, \dots, f_m$ is an orthonormal basis of W , we know how to write Te_k as a linear combination of the f_j 's [see 6.30(a)]:

$$Te_k = \langle Te_k, f_1 \rangle f_1 + \dots + \langle Te_k, f_m \rangle f_m.$$

Thus

the entry in row j , column k , of $\mathcal{M}(T)$ is $\langle Te_k, f_j \rangle$.

In the statement above, replace T with T^* and interchange $e_1, \dots, e_n$ and $f_1, \dots, f_m$. This shows that the entry in row j , column k , of $\mathcal{M}(T^*)$ is $\langle T^* f_k, e_j \rangle$, which equals $\langle f_k, Te_j \rangle$, which equals $\overline{\langle Te_j, f_k \rangle}$, which equals the complex conjugate of the entry in row k , column j , of $\mathcal{M}(T)$. Thus $\mathcal{M}(T^*) = (\mathcal{M}(T))^*$. ■

The Riesz representation theorem as stated in 6.58 provides an identification of V with its dual space V' defined in 3.110. Under this identification, the orthogonal complement $U^\perp$ of a subset $U \subseteq V$ corresponds to the annihilator U^0 of U . If U is a subspace of V , then the formulas for the dimensions of $U^\perp$ and U^0 become identical under this identification—see 3.125 and 6.51.

Suppose $T: V \rightarrow W$ is a linear map. Under the identification of V with V' and the identification of W with W' , the adjoint map $T^*: W \rightarrow V$ corresponds to the dual map $T': W' \rightarrow V'$ defined in 3.118, as Exercise 32 asks you to verify. Under this identification, the formulas for null T^* and range T^* [7.6(a) and (b)] then become identical to the formulas for null T' and range T' [3.128(a) and 3.130(b)]. Furthermore, the theorem about the matrix of T^* (7.9) is analogous to the theorem about the matrix of T' (3.132).

Because orthogonal complements and adjoints are easier to deal with than annihilators and dual maps, there is no need to work with annihilators and dual maps in the context of inner product spaces.

Self-Adjoint Operators

Now we switch our attention to operators on inner product spaces. Instead of considering linear maps from V to W , we will focus on linear maps from V to V ; recall that such linear maps are called operators.

7.10 definition: *self-adjoint*

An operator $T \in \mathcal{L}(V)$ is called *self-adjoint* if $T = T^*$.

If $T \in \mathcal{L}(V)$ and $e_1, \dots, e_n$ is an orthonormal basis of V , then T is self-adjoint if and only if $\mathcal{M}(T, (e_1, \dots, e_n)) = \mathcal{M}(T, (e_1, \dots, e_n))^*$, as follows from 7.9.

7.11 example: *determining whether T is self-adjoint from its matrix*

Suppose $c \in \mathbf{F}$ and T is the operator on $\mathbf{F}^2$ whose matrix (with respect to the standard basis) is

$$\mathcal{M}(T) = \begin{pmatrix} 2 & c \\ 3 & 7 \end{pmatrix}.$$

The matrix of T^* (with respect to the standard basis) is

$$\mathcal{M}(T^*) = \begin{pmatrix} 2 & 3 \\ \bar{c} & 7 \end{pmatrix}.$$

Thus $\mathcal{M}(T) = \mathcal{M}(T^*)$ if and only if $c = 3$. Hence the operator T is self-adjoint if and only if $c = 3$.

A good analogy to keep in mind is that the adjoint on $\mathcal{L}(V)$ plays a role similar to that of the complex conjugate on $\mathbf{C}$. A complex number z is real if and only if $z = \bar{z}$; thus a self-adjoint operator ($T = T^*$) is analogous to a real number.

We will see that the analogy discussed above is reflected in some important properties of self-adjoint operators, beginning with eigenvalues in the next result.

If $\mathbf{F} = \mathbf{R}$, then by definition every eigenvalue is real, so the next result is interesting only when $\mathbf{F} = \mathbf{C}$.

An operator $T \in \mathcal{L}(V)$ is self-adjoint if and only if

$$\langle Tv, w \rangle = \langle v, Tw \rangle$$

for all $v, w \in V$.

7.12 eigenvalues of self-adjoint operators

Every eigenvalue of a self-adjoint operator is real.

Proof Suppose T is a self-adjoint operator on V . Let λ be an eigenvalue of T , and let v be a nonzero vector in V such that $Tv = \lambda v$. Then

$$\lambda \|v\|^2 = \langle \lambda v, v \rangle = \langle Tv, v \rangle = \langle v, Tv \rangle = \langle v, \lambda v \rangle = \bar{\lambda} \|v\|^2.$$

Thus $\lambda = \bar{\lambda}$, which means that λ is real, as desired. ■

The next result is false for real inner product spaces. As an example, consider the operator $T \in \mathcal{L}(\mathbf{R}^2)$ that is a counterclockwise rotation of 90° around the origin; thus $T(x, y) = (-y, x)$. Notice that Tv is orthogonal to v for every $v \in \mathbf{R}^2$, even though $T \neq 0$.

7.13 Tv is orthogonal to v for all $v \iff T = 0$ (assuming $\mathbf{F} = \mathbf{C}$)

Suppose V is a complex inner product space and $T \in \mathcal{L}(V)$. Then

$$\langle Tv, v \rangle = 0 \text{ for every } v \in V \iff T = 0.$$

Proof If $u, w \in V$, then

$$\begin{aligned} \langle Tu, w \rangle &= \frac{\langle T(u+w), u+w \rangle - \langle T(u-w), u-w \rangle}{4} \\ &\quad + \frac{\langle T(u+iw), u+iw \rangle - \langle T(u-iw), u-iw \rangle}{4} i, \end{aligned}$$

as can be verified by computing the right side. Note that each term on the right side is of the form $\langle Tv, v \rangle$ for appropriate $v \in V$.

Now suppose $\langle Tv, v \rangle = 0$ for every $v \in V$. Then the equation above implies that $\langle Tu, w \rangle = 0$ for all $u, w \in V$, which then implies that $Tu = 0$ for every $u \in V$ (take $w = Tu$). Hence $T = 0$, as desired. ■

The next result is false for real inner product spaces, as shown by considering any operator on a real inner product space that is not self-adjoint.

The next result provides another good example of how self-adjoint operators behave like real numbers.

7.14 $\langle Tv, v \rangle$ is real for all $v \iff T$ is self-adjoint (assuming $\mathbf{F} = \mathbf{C}$)

Suppose V is a complex inner product space and $T \in \mathcal{L}(V)$. Then

$$T \text{ is self-adjoint} \iff \langle Tv, v \rangle \in \mathbf{R} \text{ for every } v \in V.$$

Proof If $v \in V$, then

$$7.15 \quad \langle T^*v, v \rangle = \overline{\langle v, T^*v \rangle} = \overline{\langle Tv, v \rangle}.$$

Now

$$\begin{aligned} T \text{ is self-adjoint} &\iff T - T^* = 0 \\ &\iff \langle (T - T^*)v, v \rangle = 0 \text{ for every } v \in V \\ &\iff \langle Tv, v \rangle - \overline{\langle Tv, v \rangle} = 0 \text{ for every } v \in V \\ &\iff \langle Tv, v \rangle \in \mathbf{R} \text{ for every } v \in V, \end{aligned}$$

where the second equivalence follows from 7.13 as applied to $T - T^*$ and the third equivalence follows from 7.15. ■

On a real inner product space V , a nonzero operator T might satisfy $\langle Tv, v \rangle = 0$ for all $v \in V$. However, the next result shows that this cannot happen for a self-adjoint operator.

7.16 T self-adjoint and $\langle Tv, v \rangle = 0$ for all $v \iff T = 0$

Suppose T is a self-adjoint operator on V . Then

$$\langle Tv, v \rangle = 0 \text{ for every } v \in V \iff T = 0.$$

Proof We have already proved this (without the hypothesis that T is self-adjoint) when V is a complex inner product space (see 7.13). Thus we can assume that V is a real inner product space. If $u, w \in V$, then

$$7.17 \quad \langle Tu, w \rangle = \frac{\langle T(u+w), u+w \rangle - \langle T(u-w), u-w \rangle}{4},$$

as can be proved by computing the right side using the equation

$$\langle Tw, u \rangle = \langle w, Tu \rangle = \langle Tu, w \rangle,$$

where the first equality holds because T is self-adjoint and the second equality holds because we are working in a real inner product space.

Now suppose $\langle Tv, v \rangle = 0$ for every $v \in V$. Because each term on the right side of 7.17 is of the form $\langle Tv, v \rangle$ for appropriate v , this implies that $\langle Tu, w \rangle = 0$ for all $u, w \in V$. This implies that $Tu = 0$ for every $u \in V$ (take $w = Tu$). Hence $T = 0$, as desired. ■

Normal Operators

7.18 definition: *normal*

- An operator on an inner product space is called *normal* if it commutes with its adjoint.
- In other words, $T \in \mathcal{L}(V)$ is normal if $TT^* = T^*T$.

Every self-adjoint operator is normal, because if T is self-adjoint then $T^* = T$ and hence T commutes with T^* .

7.19 example: *an operator that is normal but not self-adjoint*

Let T be the operator on $\mathbb{F}^2$ whose matrix (with respect to the standard basis) is

$$\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}.$$

Thus $T(w, z) = (2w - 3z, 3w + 2z)$.

This operator T is not self-adjoint because the entry in row 2, column 1 (which equals 3) does not equal the complex conjugate of the entry in row 1, column 2 (which equals -3).

The matrix of TT^* equals

$$\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ -3 & 2 \end{pmatrix}, \text{ which equals } \begin{pmatrix} 13 & 0 \\ 0 & 13 \end{pmatrix}.$$

Similarly, the matrix of T^*T equals

$$\begin{pmatrix} 2 & 3 \\ -3 & 2 \end{pmatrix} \begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}, \text{ which equals } \begin{pmatrix} 13 & 0 \\ 0 & 13 \end{pmatrix}.$$

Because TT^* and T^*T have the same matrix, we see that $TT^* = T^*T$. Thus T is normal.

In the next section we will see why normal operators are worthy of special attention. The next result provides a useful characterization of normal operators.

7.20 *T is normal if and only if Tv and T^*v have the same norm*

Suppose $T \in \mathcal{L}(V)$. Then

$$T \text{ is normal} \iff \|Tv\| = \|T^*v\| \text{ for every } v \in V.$$

Proof We have

$$\begin{aligned} T \text{ is normal} &\iff T^*T - TT^* = 0 \\ &\iff \langle (T^*T - TT^*)v, v \rangle = 0 \text{ for every } v \in V \\ &\iff \langle T^*Tv, v \rangle = \langle TT^*v, v \rangle \text{ for every } v \in V \\ &\iff \langle Tv, Tv \rangle = \langle T^*v, T^*v \rangle \text{ for every } v \in V \\ &\iff \|Tv\|^2 = \|T^*v\|^2 \text{ for every } v \in V \\ &\iff \|Tv\| = \|T^*v\| \text{ for every } v \in V, \end{aligned}$$

where we used 7.16 to establish the second equivalence (note that the operator $T^*T - TT^*$ is self-adjoint). ■

The next result presents several consequences of the result above. Compare (e) of the next result to Exercise 3. That exercise states that the eigenvalues of the adjoint of each operator are equal (as a set) to the complex conjugates of the eigenvalues of the operator. The exercise says nothing about eigenvectors, because an operator and its adjoint may have different eigenvectors. However, (e) of the next result implies that a normal operator and its adjoint have the same eigenvectors.

7.21 range, null space, and eigenvectors of a normal operator

Suppose $T \in \mathcal{L}(V)$ is normal. Then

- (a) $\text{null } T = \text{null } T^*$;
- (b) $\text{range } T = \text{range } T^*$;
- (c) $V = \text{null } T \oplus \text{range } T$;
- (d) $T - \lambda I$ is normal for every $\lambda \in \mathbb{F}$;
- (e) if $v \in V$ and $\lambda \in \mathbb{F}$, then $Tv = \lambda v$ if and only if $T^*v = \bar{\lambda}v$.

Proof

- (a) Suppose $v \in V$. Then

$$v \in \text{null } T \iff \|Tv\| = 0 \iff \|T^*v\| = 0 \iff v \in \text{null } T^*,$$

where the middle equivalence above follows from 7.20. Thus $\text{null } T = \text{null } T^*$.

- (b) We have

$$\text{range } T = (\text{null } T^*)^\perp = (\text{null } T)^\perp = \text{range } T^*,$$

where the first equality comes from 7.6(d), the second equality comes from (a) in this result, and the third equality comes from 7.6(b).

- (c) We have

$$V = (\text{null } T) \oplus (\text{null } T)^\perp = \text{null } T \oplus \text{range } T^* = \text{null } T \oplus \text{range } T,$$

where the first equality comes from 6.49, the second equality comes from 7.6(b), and the third equality comes from (b) in this result.

- (d) Suppose $\lambda \in \mathbb{F}$. Then

$$\begin{aligned} (T - \lambda I)(T - \lambda I)^* &= (T - \lambda I)(T^* - \bar{\lambda}I) \\ &= TT^* - \bar{\lambda}T - \lambda T^* + |\lambda|^2 I \\ &= T^*T - \bar{\lambda}T - \lambda T^* + |\lambda|^2 I \\ &= (T^* - \bar{\lambda}I)(T - \lambda I) \\ &= (T - \lambda I)^*(T - \lambda I). \end{aligned}$$

Thus $T - \lambda I$ commutes with its adjoint. Hence $T - \lambda I$ is normal.

- (e) Suppose $v \in V$ and $\lambda \in \mathbb{F}$. Then (d) and 7.20 imply that

$$\|(T - \lambda I)v\| = \|(T - \lambda I)^*v\| = \|(T^* - \bar{\lambda}I)v\|.$$

Thus $\|(T - \lambda I)v\| = 0$ if and only if $\|(T^* - \bar{\lambda}I)v\| = 0$. Hence $Tv = \lambda v$ if and only if $T^*v = \bar{\lambda}v$. ■

Because every self-adjoint operator is normal, the next result applies in particular to self-adjoint operators.

7.22 *orthogonal eigenvectors for normal operators*

Suppose $T \in \mathcal{L}(V)$ is normal. Then eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

Proof Suppose α, β are distinct eigenvalues of T , with corresponding eigenvectors u, v . Thus $Tu = \alpha u$ and $Tv = \beta v$. From 7.21(e) we have $T^*v = \bar{\beta}v$. Thus

$$\begin{aligned}(\alpha - \beta)\langle u, v \rangle &= \langle \alpha u, v \rangle - \langle u, \bar{\beta}v \rangle \\&= \langle Tu, v \rangle - \langle u, T^*v \rangle \\&= 0.\end{aligned}$$

Because $\alpha \neq \beta$, the equation above implies that $\langle u, v \rangle = 0$. Thus u and v are orthogonal, as desired. ■

As stated here, the next result makes sense only when $\mathbf{F} = \mathbf{C}$. However, see Exercise 12 for a version that makes sense when $\mathbf{F} = \mathbf{C}$ and when $\mathbf{F} = \mathbf{R}$.

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Under the analogy between $\mathcal{L}(V)$ and $\mathbf{C}$, with the adjoint on $\mathcal{L}(V)$ playing a similar role to that of the complex conjugate on $\mathbf{C}$, the operators A and B as defined by 7.24 correspond to the real and imaginary parts of T . Thus the informal title of the result below should make sense.

7.23 *T is normal $\iff$ the real and imaginary parts of T commute*

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then T is normal if and only if there exist commuting self-adjoint operators A and B such that $T = A + iB$.

Proof First suppose T is normal. Let

$$7.24 \quad A = \frac{T + T^*}{2} \quad \text{and} \quad B = \frac{T - T^*}{2i}.$$

Then A and B are self-adjoint and $T = A + iB$. A quick computation shows that

$$7.25 \quad AB - BA = \frac{T^*T - TT^*}{2i}.$$

Because T is normal, the right side of the equation above equals 0. Thus the operators A and B commute, as desired.

To prove the implication in the other direction, now suppose there exist commuting self-adjoint operators A and B such that $T = A + iB$. Then $T^* = A - iB$. Adding the last two equations and then dividing by 2 produces the equation for A in 7.24. Subtracting the last two equations and then dividing by $2i$ produces the equation for B in 7.24. Now 7.24 implies 7.25. Because B and A commute, 7.25 implies that T is normal, as desired. ■

Exercises 7A

- 1 Suppose n is a positive integer. Define $T \in \mathcal{L}(\mathbf{F}^n)$ by

$$T(z_1, \dots, z_n) = (0, z_1, \dots, z_{n-1}).$$

Find a formula for $T^*(z_1, \dots, z_n)$.

- 2 Suppose $T \in \mathcal{L}(V, W)$. Prove that

$$T = 0 \iff T^* = 0 \iff T^*T = 0 \iff TT^* = 0.$$

- 3 Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbf{F}$. Prove that

$$\lambda \text{ is an eigenvalue of } T \iff \bar{\lambda} \text{ is an eigenvalue of } T^*.$$

- 4 Suppose $T \in \mathcal{L}(V)$ and U is a subspace of V . Prove that

$$U \text{ is invariant under } T \iff U^\perp \text{ is invariant under } T^*.$$

- 5 Suppose $T \in \mathcal{L}(V, W)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $f_1, \dots, f_m$ is an orthonormal basis of W . Prove that

$$\|Te_1\|^2 + \dots + \|Te_n\|^2 = \|T^*f_1\|^2 + \dots + \|T^*f_m\|^2.$$

The numbers $\|Te_1\|^2, \dots, \|Te_n\|^2$ in the equation above depend on the orthonormal basis $e_1, \dots, e_n$, but the right side of the equation does not depend on $e_1, \dots, e_n$. Thus the equation above shows that the sum on the left side does not depend on which orthonormal basis $e_1, \dots, e_n$ is used.

- 6 Suppose $T \in \mathcal{L}(V, W)$. Prove that
 (a) T is injective $\iff T^*$ is surjective;
 (b) T is surjective $\iff T^*$ is injective.

- 7 Prove that if $T \in \mathcal{L}(V, W)$, then
 (a) $\dim \text{null } T^* = \dim \text{null } T + \dim W - \dim V$;
 (b) $\dim \text{range } T^* = \dim \text{range } T$.

- 8 Suppose A is an m -by- n matrix with entries in $\mathbf{F}$. Use (b) in Exercise 7 to prove that the row rank of A equals the column rank of A .

This exercise asks for yet another alternative proof of a result that was previously proved in 3.57 and 3.133.

- 9 Prove that the product of two self-adjoint operators on V is self-adjoint if and only if the two operators commute.
- 10 Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that T is self-adjoint if and only if

$$\langle Tv, v \rangle = \langle T^*v, v \rangle$$

for all $v \in V$.

- 11 Define an operator $S: \mathbf{F}^2 \rightarrow \mathbf{F}^2$ by $S(w, z) = (-z, w)$.
- Find a formula for S^* .
 - Show that S is normal but not self-adjoint.
 - Find all eigenvalues of S .

If $\mathbf{F} = \mathbf{R}$, then S is the operator on $\mathbf{R}^2$ of counterclockwise rotation by 90° .

- 12 An operator $B \in \mathcal{L}(V)$ is called *skew* if

$$B^* = -B.$$

Suppose that $T \in \mathcal{L}(V)$. Prove that T is normal if and only if there exist commuting operators A and B such that A is self-adjoint, B is a skew operator, and $T = A + B$.

- 13 Suppose $\mathbf{F} = \mathbf{R}$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by $\mathcal{A}T = T^*$ for all $T \in \mathcal{L}(V)$.
- Find all eigenvalues of $\mathcal{A}$.
 - Find the minimal polynomial of $\mathcal{A}$.
- 14 Define an inner product on $\mathcal{P}_2(\mathbf{R})$ by $\langle p, q \rangle = \int_0^1 pq$. Define an operator $T \in \mathcal{L}(\mathcal{P}_2(\mathbf{R}))$ by
- $$T(ax^2 + bx + c) = bx.$$
- Show that with this inner product, the operator T is not self-adjoint.
 - The matrix of T with respect to the basis $1, x, x^2$ is

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

This matrix equals its conjugate transpose, even though T is not self-adjoint. Explain why this is not a contradiction.

- 15 Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that
- T is self-adjoint $\iff T^{-1}$ is self-adjoint;
 - T is normal $\iff T^{-1}$ is normal.
- 16 Suppose $\mathbf{F} = \mathbf{R}$.
- Show that the set of self-adjoint operators on V is a subspace of $\mathcal{L}(V)$.
 - What is the dimension of the subspace of $\mathcal{L}(V)$ in (a) [in terms of $\dim V$]?
- 17 Suppose $\mathbf{F} = \mathbf{C}$. Show that the set of self-adjoint operators on V is not a subspace of $\mathcal{L}(V)$.
- 18 Suppose $\dim V \geq 2$. Show that the set of normal operators on V is not a subspace of $\mathcal{L}(V)$.

- 19** Suppose $T \in \mathcal{L}(V)$ and $\|T^*v\| \leq \|Tv\|$ for every $v \in V$. Prove that T is normal.

This exercise fails on infinite-dimensional inner product spaces, leading to what are called hyponormal operators, which have a well-developed theory.

- 20** Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that the following are equivalent.
- (a) P is self-adjoint.
 - (b) P is normal.
 - (c) There is a subspace U of V such that $P = P_U$.

- 21** Suppose $D: \mathcal{P}_8(\mathbf{R}) \rightarrow \mathcal{P}_8(\mathbf{R})$ is the differentiation operator defined by $Dp = p'$. Prove that there does not exist an inner product on $\mathcal{P}_8(\mathbf{R})$ that makes D a normal operator.

- 22** Give an example of an operator $T \in \mathcal{L}(\mathbf{R}^3)$ such that T is normal but not self-adjoint.

- 23** Suppose T is a normal operator on V . Suppose also that $v, w \in V$ satisfy the equations

$$\|v\| = \|w\| = 2, \quad Tv = 3v, \quad Tw = 4w.$$

Show that $\|T(v + w)\| = 10$.

- 24** Suppose $T \in \mathcal{L}(V)$ and

$$a_0 + a_1z + a_2z^2 + \cdots + a_{m-1}z^{m-1} + z^m$$

is the minimal polynomial of T . Prove that the minimal polynomial of T^* is

$$\overline{a_0} + \overline{a_1}z + \overline{a_2}z^2 + \cdots + \overline{a_{m-1}}z^{m-1} + z^m.$$

This exercise shows that the minimal polynomial of T^ equals the minimal polynomial of T if $\mathbf{F} = \mathbf{R}$.*

- 25** Suppose $T \in \mathcal{L}(V)$. Prove that T is diagonalizable if and only if T^* is diagonalizable.

- 26** Fix $u, x \in V$. Define $T \in \mathcal{L}(V)$ by $Tv = \langle v, u \rangle x$ for every $v \in V$.

- (a) Prove that if V is a real vector space, then T is self-adjoint if and only if the list u, x is linearly dependent.
- (b) Prove that T is normal if and only if the list u, x is linearly dependent.

- 27** Suppose $T \in \mathcal{L}(V)$ is normal. Prove that

$$\text{null } T^k = \text{null } T \quad \text{and} \quad \text{range } T^k = \text{range } T$$

for every positive integer k .

- 28** Suppose $T \in \mathcal{L}(V)$ is normal. Prove that if $\lambda \in \mathbf{F}$, then the minimal polynomial of T is not a polynomial multiple of $(x - \lambda)^2$.

- 29** Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and there is an orthonormal basis $e_1, \dots, e_n$ of V such that $\|Te_k\| = \|T^*e_k\|$ for each $k = 1, \dots, n$, then T is normal.
- 30** Suppose that $T \in \mathcal{L}(\mathbb{F}^3)$ is normal and $T(1, 1, 1) = (2, 2, 2)$. Suppose $(z_1, z_2, z_3) \in \text{null } T$. Prove that $z_1 + z_2 + z_3 = 0$.
- 31** Fix a positive integer n . In the inner product space of continuous real-valued functions on $[-\pi, \pi]$ with inner product $\langle f, g \rangle = \int_{-\pi}^{\pi} fg$, let
- $$V = \text{span}(1, \cos x, \cos 2x, \dots, \cos nx, \sin x, \sin 2x, \dots, \sin nx).$$
- (a) Define $D \in \mathcal{L}(V)$ by $Df = f'$. Show that $D^* = -D$. Conclude that D is normal but not self-adjoint.
- (b) Define $T \in \mathcal{L}(V)$ by $Tf = f''$. Show that T is self-adjoint.
- 32** Suppose $T: V \rightarrow W$ is a linear map. Show that under the standard identification of V with V' (see 6.58) and the corresponding identification of W with W' , the adjoint map $T^*: W \rightarrow V$ corresponds to the dual map $T': W' \rightarrow V'$. More precisely, show that

$$T'(\varphi_w) = \varphi_{T^*w}$$

for all $w \in W$, where φ_w and φ_{T^*w} are defined as in 6.58.

7B Spectral Theorem

Recall that a diagonal matrix is a square matrix that is 0 everywhere except possibly on the diagonal. Recall that an operator on V is called diagonalizable if the operator has a diagonal matrix with respect to some basis of V . Recall also that this happens if and only if there is a basis of V consisting of eigenvectors of the operator (see 5.55).

The nicest operators on V are those for which there is an *orthonormal* basis of V with respect to which the operator has a diagonal matrix. These are precisely the operators $T \in \mathcal{L}(V)$ such that there is an orthonormal basis of V consisting of eigenvectors of T . Our goal in this section is to prove the spectral theorem, which characterizes these operators as the self-adjoint operators when $\mathbf{F} = \mathbf{R}$ and as the normal operators when $\mathbf{F} = \mathbf{C}$.

The spectral theorem is probably the most useful tool in the study of operators on inner product spaces. Its extension to certain infinite-dimensional inner product spaces (see, for example, Section 10D of the author's book *Measure, Integration & Real Analysis*) plays a key role in functional analysis.

Because the conclusion of the spectral theorem depends on $\mathbf{F}$, we will break the spectral theorem into two pieces, called the real spectral theorem and the complex spectral theorem.

Real Spectral Theorem

To prove the real spectral theorem, we will need two preliminary results. These preliminary results hold on both real and complex inner product spaces, but they are not needed for the proof of the complex spectral theorem.

You could guess that the next result is true and even discover its proof by thinking about quadratic polynomials with real coefficients. Specifically, suppose $b, c \in \mathbf{R}$ and $b^2 < 4c$. Let x be a real number. Then

This completing-the-square technique can be used to derive the quadratic formula.

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right) > 0.$$

In particular, $x^2 + bx + c$ is an invertible real number (a convoluted way of saying that it is not 0). Replacing the real number x with a self-adjoint operator (recall the analogy between real numbers and self-adjoint operators) leads to the next result.

7.26 invertible quadratic expressions

Suppose $T \in \mathcal{L}(V)$ is self-adjoint and $b, c \in \mathbf{R}$ are such that $b^2 < 4c$. Then

$$T^2 + bT + cI$$

is an invertible operator.

Proof Let v be a nonzero vector in V . Then

$$\begin{aligned}
 \langle (T^2 + bT + cI)v, v \rangle &= \langle T^2v, v \rangle + b\langle Tv, v \rangle + c\langle v, v \rangle \\
 &= \langle Tv, Tv \rangle + b\langle Tv, v \rangle + c\|v\|^2 \\
 &\geq \|Tv\|^2 - |b| \|Tv\| \|v\| + c\|v\|^2 \\
 &= \left(\|Tv\| - \frac{|b| \|v\|}{2} \right)^2 + \left(c - \frac{b^2}{4} \right) \|v\|^2 \\
 &> 0,
 \end{aligned}$$

where the third line above holds by the Cauchy–Schwarz inequality (6.14). The last inequality implies that $(T^2 + bT + cI)v \neq 0$. Thus $T^2 + bT + cI$ is injective, which implies that it is invertible (see 3.65). ■

The next result will be a key tool in our proof of the real spectral theorem.

7.27 minimal polynomial of self-adjoint operator

Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Then the minimal polynomial of T equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some $\lambda_1, \dots, \lambda_m \in \mathbf{R}$.

Proof First suppose $\mathbf{F} = \mathbf{C}$. The zeros of the minimal polynomial of T are the eigenvalues of T [by 5.27(a)]. All eigenvalues of T are real (by 7.12). Thus the second version of the fundamental theorem of algebra (see 4.13) tells us that the minimal polynomial of T has the desired form.

Now suppose $\mathbf{F} = \mathbf{R}$. By the factorization of a polynomial over $\mathbf{R}$ (see 4.16) there exist $\lambda_1, \dots, \lambda_m \in \mathbf{R}$ and $b_1, \dots, b_N, c_1, \dots, c_N \in \mathbf{R}$ with $b_k^2 < 4c_k$ for each k such that the minimal polynomial of T equals

$$7.28 \quad (z - \lambda_1) \cdots (z - \lambda_m)(z^2 + b_1z + c_1) \cdots (z^2 + b_Nz + c_N);$$

here either m or N might equal 0, meaning that there are no terms of the corresponding form. Now

$$(T - \lambda_1 I) \cdots (T - \lambda_m I)(T^2 + b_1T + c_1I) \cdots (T^2 + b_NT + c_NI) = 0.$$

If $N > 0$, then we could multiply both sides of the equation above on the right by the inverse of $T^2 + b_NT + c_NI$ (which is an invertible operator by 7.26) to obtain a polynomial expression of T that equals 0. The corresponding polynomial would have degree two less than the degree of 7.28, violating the minimality of the degree of the polynomial with this property. Thus we must have $N = 0$, which means that the minimal polynomial in 7.28 has the form $(z - \lambda_1) \cdots (z - \lambda_m)$, as desired. ■

The result above along with 5.27(a) implies that every self-adjoint operator has an eigenvalue. In fact, as we will see in the next result, self-adjoint operators have enough eigenvectors to form a basis.

The next result, which gives a complete description of the self-adjoint operators on a real inner product space, is one of the major theorems in linear algebra.

7.29 real spectral theorem

Suppose $\mathbf{F} = \mathbf{R}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent.

- (a) T is self-adjoint.
- (b) T has a diagonal matrix with respect to some orthonormal basis of V .
- (c) V has an orthonormal basis consisting of eigenvectors of T .

Proof First suppose (a) holds, so T is self-adjoint. Our results on minimal polynomials, specifically 6.37 and 7.27, imply that T has an upper-triangular matrix with respect to some orthonormal basis of V . With respect to this orthonormal basis, the matrix of T^* is the transpose of the matrix of T . However, $T^* = T$. Thus the transpose of the matrix of T equals the matrix of T . Because the matrix of T is upper-triangular, this means that all entries of the matrix above and below the diagonal are 0. Hence the matrix of T is a diagonal matrix with respect to the orthonormal basis. Thus (a) implies (b).

Conversely, now suppose (b) holds, so T has a diagonal matrix with respect to some orthonormal basis of V . That diagonal matrix equals its transpose. Thus with respect to that basis, the matrix of T^* equals the matrix of T . Hence $T^* = T$, proving that (b) implies (a).

The equivalence of (b) and (c) follows from the definitions [or see the proof that (a) and (b) are equivalent in 5.55]. ■

7.30 example: an orthonormal basis of eigenvectors for an operator

Consider the operator T on $\mathbf{R}^3$ whose matrix (with respect to the standard basis) is

$$\begin{pmatrix} 14 & -13 & 8 \\ -13 & 14 & 8 \\ 8 & 8 & -7 \end{pmatrix}.$$

This matrix with real entries equals its transpose; thus T is self-adjoint. As you can verify,

$$\frac{(1, -1, 0)}{\sqrt{2}}, \frac{(1, 1, 1)}{\sqrt{3}}, \frac{(1, 1, -2)}{\sqrt{6}}$$

is an orthonormal basis of $\mathbf{R}^3$ consisting of eigenvectors of T . With respect to this basis, the matrix of T is the diagonal matrix

$$\begin{pmatrix} 27 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & -15 \end{pmatrix}.$$

See Exercise 17 for a version of the real spectral theorem that applies simultaneously to more than one operator.

Complex Spectral Theorem

The next result gives a complete description of the normal operators on a complex inner product space.

7.31 complex spectral theorem

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent.

- (a) T is normal.
- (b) T has a diagonal matrix with respect to some orthonormal basis of V .
- (c) V has an orthonormal basis consisting of eigenvectors of T .

Proof First suppose (a) holds, so T is normal. By Schur's theorem (6.38), there is an orthonormal basis $e_1, \dots, e_n$ of V with respect to which T has an upper-triangular matrix. Thus we can write

$$7.32 \quad \mathcal{M}(T, (e_1, \dots, e_n)) = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ & \ddots & \vdots \\ 0 & & a_{n,n} \end{pmatrix}.$$

We will show that this matrix is actually a diagonal matrix.

We see from the matrix above that

$$\begin{aligned} \|Te_1\|^2 &= |a_{1,1}|^2, \\ \|T^*e_1\|^2 &= |a_{1,1}|^2 + |a_{1,2}|^2 + \cdots + |a_{1,n}|^2. \end{aligned}$$

Because T is normal, $\|Te_1\| = \|T^*e_1\|$ (see 7.20). Thus the two equations above imply that all entries in the first row of the matrix in 7.32, except possibly the first entry $a_{1,1}$, equal 0.

Now 7.32 implies

$$\|Te_2\|^2 = |a_{2,2}|^2$$

(because $a_{1,2} = 0$, as we showed in the paragraph above) and

$$\|T^*e_2\|^2 = |a_{2,2}|^2 + |a_{2,3}|^2 + \cdots + |a_{2,n}|^2.$$

Because T is normal, $\|Te_2\| = \|T^*e_2\|$. Thus the two equations above imply that all entries in the second row of the matrix in 7.32, except possibly the diagonal entry $a_{2,2}$, equal 0.

Continuing in this fashion, we see that all nondiagonal entries in the matrix 7.32 equal 0. Thus (b) holds, completing the proof that (a) implies (b).

Now suppose (b) holds, so T has a diagonal matrix with respect to some orthonormal basis of V . The matrix of T^* (with respect to the same basis) is obtained by taking the conjugate transpose of the matrix of T ; hence T^* also has a diagonal matrix. Any two diagonal matrices commute; thus T commutes with T^* , which means that T is normal. In other words, (a) holds, completing the proof that (b) implies (a).

The equivalence of (b) and (c) follows from the definitions (also see 5.55). ■

See Exercises 13 and 20 for alternative proofs that (a) implies (b) in the previous result.

Exercises 14 and 15 interpret the real spectral theorem and the complex spectral theorem by expressing the domain space as an orthogonal direct sum of eigenspaces.

See Exercise 16 for a version of the complex spectral theorem that applies simultaneously to more than one operator.

The main conclusion of the complex spectral theorem is that every normal operator on a complex finite-dimensional inner product space is diagonalizable by an orthonormal basis, as illustrated by the next example.

7.33 example: an orthonormal basis of eigenvectors for an operator

Consider the operator $T \in \mathcal{L}(\mathbf{C}^2)$ defined by $T(w, z) = (2w - 3z, 3w + 2z)$. The matrix of T (with respect to the standard basis) is

$$\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}.$$

As we saw in Example 7.19, T is a normal operator.

As you can verify,

$$\frac{1}{\sqrt{2}}(i, 1), \frac{1}{\sqrt{2}}(-i, 1)$$

is an orthonormal basis of $\mathbf{C}^2$ consisting of eigenvectors of T , and with respect to this basis the matrix of T is the diagonal matrix

$$\begin{pmatrix} 2 + 3i & 0 \\ 0 & 2 - 3i \end{pmatrix}.$$

Exercises 7B

- 1 Prove that a normal operator on a complex inner product space is self-adjoint if and only if all its eigenvalues are real.

This exercise strengthens the analogy (for normal operators) between self-adjoint operators and real numbers.

- 2 Suppose $\mathbf{F} = \mathbf{C}$. Suppose $T \in \mathcal{L}(V)$ is normal and has only one eigenvalue. Prove that T is a scalar multiple of the identity operator.
- 3 Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is normal. Prove that the set of eigenvalues of T is contained in $\{0, 1\}$ if and only if there is a subspace U of V such that $T = P_U$.
- 4 Prove that a normal operator on a complex inner product space is skew (meaning it equals the negative of its adjoint) if and only if all its eigenvalues are purely imaginary (meaning that they have real part equal to 0).

- 5 Prove or give a counterexample: If $T \in \mathcal{L}(\mathbb{C}^3)$ is a diagonalizable operator, then T is normal (with respect to the usual inner product).
- 6 Suppose V is a complex inner product space and $T \in \mathcal{L}(V)$ is a normal operator such that $T^9 = T^8$. Prove that T is self-adjoint and $T^2 = T$.
- 7 Give an example of an operator T on a complex vector space such that $T^9 = T^8$ but $T^2 \neq T$.
- 8 Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that T is normal if and only if every eigenvector of T is also an eigenvector of T^* .
- 9 Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that T is normal if and only if there exists a polynomial $p \in \mathcal{P}(\mathbf{C})$ such that $T^* = p(T)$.
- 10 Suppose V is a complex inner product space. Prove that every normal operator on V has a square root.

*An operator $S \in \mathcal{L}(V)$ is called a **square root** of $T \in \mathcal{L}(V)$ if $S^2 = T$. We will discuss more about square roots of operators in Sections 7C and 8C.*

- 11 Prove that every self-adjoint operator on V has a cube root.
*An operator $S \in \mathcal{L}(V)$ is called a **cube root** of $T \in \mathcal{L}(V)$ if $S^3 = T$.*
- 12 Suppose V is a complex vector space and $T \in \mathcal{L}(V)$ is normal. Prove that if S is an operator on V that commutes with T , then S commutes with T^* .
The result in this exercise is called Fuglede's theorem.

- 13 Without using the complex spectral theorem, use the version of Schur's theorem that applies to two commuting operators (take $\mathcal{E} = \{T, T^*\}$ in Exercise 20 in Section 6B) to give a different proof that if $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$ is normal, then T has a diagonal matrix with respect to some orthonormal basis of V .
- 14 Suppose $\mathbf{F} = \mathbf{R}$ and $T \in \mathcal{L}(V)$. Prove that T is self-adjoint if and only if all pairs of eigenvectors corresponding to distinct eigenvalues of T are orthogonal and $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$, where $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T .
- 15 Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that T is normal if and only if all pairs of eigenvectors corresponding to distinct eigenvalues of T are orthogonal and $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$, where $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T .
- 16 Suppose $\mathbf{F} = \mathbf{C}$ and $\mathcal{E} \subseteq \mathcal{L}(V)$. Prove that there is an orthonormal basis of V with respect to which every element of $\mathcal{E}$ has a diagonal matrix if and only if S and T are commuting normal operators for all $S, T \in \mathcal{E}$.

This exercise extends the complex spectral theorem to the context of a collection of commuting normal operators.

- 17** Suppose $\mathbf{F} = \mathbf{R}$ and $\mathcal{E} \subseteq \mathcal{L}(V)$. Prove that there is an orthonormal basis of V with respect to which every element of $\mathcal{E}$ has a diagonal matrix if and only if S and T are commuting self-adjoint operators for all $S, T \in \mathcal{E}$.

This exercise extends the real spectral theorem to the context of a collection of commuting self-adjoint operators.

- 18** Give an example of a real inner product space V , an operator $T \in \mathcal{L}(V)$, and real numbers b, c with $b^2 < 4c$ such that

$$T^2 + bT + cI$$

is not invertible.

This exercise shows that the hypothesis that T is self-adjoint cannot be deleted in 7.26, even for real vector spaces.

- 19** Suppose $T \in \mathcal{L}(V)$ is self-adjoint and U is a subspace of V that is invariant under T .

- (a) Prove that $U^\perp$ is invariant under T .
- (b) Prove that $T|_U \in \mathcal{L}(U)$ is self-adjoint.
- (c) Prove that $T|_{U^\perp} \in \mathcal{L}(U^\perp)$ is self-adjoint.

- 20** Suppose $T \in \mathcal{L}(V)$ is normal and U is a subspace of V that is invariant under T .

- (a) Prove that $U^\perp$ is invariant under T .
- (b) Prove that U is invariant under T^* .
- (c) Prove that $(T|_U)^* = (T^*)|_U$.
- (d) Prove that $T|_U \in \mathcal{L}(U)$ and $T|_{U^\perp} \in \mathcal{L}(U^\perp)$ are normal operators.

This exercise can be used to give yet another proof of the complex spectral theorem (use induction on $\dim V$ and the result that T has an eigenvector).

- 21** Suppose that T is a self-adjoint operator on a finite-dimensional inner product space and that 2 and 3 are the only eigenvalues of T . Prove that

$$T^2 - 5T + 6I = 0.$$

- 22** Give an example of an operator $T \in \mathcal{L}(\mathbf{C}^3)$ such that 2 and 3 are the only eigenvalues of T and $T^2 - 5T + 6I \neq 0$.

- 23** Suppose $T \in \mathcal{L}(V)$ is self-adjoint, $\lambda \in \mathbf{F}$, and $\epsilon > 0$. Suppose there exists $v \in V$ such that $\|v\| = 1$ and

$$\|Tv - \lambda v\| < \epsilon.$$

Prove that T has an eigenvalue λ' such that $|\lambda - \lambda'| < \epsilon$.

This exercise shows that for a self-adjoint operator, a number that is close to satisfying an equation that would make it an eigenvalue is close to an eigenvalue.

- 24** Suppose U is a finite-dimensional vector space and $T \in \mathcal{L}(U)$.
- (a) Suppose $\mathbf{F} = \mathbf{R}$. Prove that T is diagonalizable if and only if there is a basis of U such that the matrix of T with respect to this basis equals its transpose.
 - (b) Suppose $\mathbf{F} = \mathbf{C}$. Prove that T is diagonalizable if and only if there is a basis of U such that the matrix of T with respect to this basis commutes with its conjugate transpose.

This exercise adds another equivalence to the list of conditions equivalent to diagonalizability in 5.55.

- 25** Suppose that $T \in \mathcal{L}(V)$ and there is an orthonormal basis $e_1, \dots, e_n$ of V consisting of eigenvectors of T , with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$. Show that if $k \in \{1, \dots, n\}$, then the pseudoinverse $T^\dagger$ satisfies the equation

$$T^\dagger e_k = \begin{cases} \frac{1}{\lambda_k} e_k & \text{if } \lambda_k \neq 0, \\ 0 & \text{if } \lambda_k = 0. \end{cases}$$

7C Positive Operators

7.34 definition: *positive operator*

An operator $T \in \mathcal{L}(V)$ is called *positive* if T is self-adjoint and

$$\langle Tv, v \rangle \geq 0$$

for all $v \in V$.

If V is a complex vector space, then the requirement that T be self-adjoint can be dropped from the definition above (by 7.14).

7.35 example: *positive operators*

- Let $T \in \mathcal{L}(\mathbb{F}^2)$ be the operator whose matrix (using the standard basis) is $\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$. Then T is self-adjoint and $\langle T(w, z), (w, z) \rangle = 2|w|^2 - 2\operatorname{Re}(w\bar{z}) + |z|^2 = |w - z|^2 + |w|^2 \geq 0$ for all $(w, z) \in \mathbb{F}^2$. Thus T is a positive operator.
- If U is a subspace of V , then the orthogonal projection P_U is a positive operator, as you should verify.
- If $T \in \mathcal{L}(V)$ is self-adjoint and $b, c \in \mathbb{R}$ are such that $b^2 < 4c$, then $T^2 + bT + cI$ is a positive operator, as shown by the proof of 7.26.

7.36 definition: *square root*

An operator R is called a *square root* of an operator T if $R^2 = T$.

7.37 example: *square root of an operator*

If $T \in \mathcal{L}(\mathbb{F}^3)$ is defined by $T(z_1, z_2, z_3) = (z_3, 0, 0)$, then the operator $R \in \mathcal{L}(\mathbb{F}^3)$ defined by $R(z_1, z_2, z_3) = (z_2, z_3, 0)$ is a square root of T because $R^2 = T$, as you can verify.

The characterizations of the positive operators in the next result correspond to characterizations of the nonnegative numbers among $\mathbb{C}$. Specifically, a number $z \in \mathbb{C}$ is nonnegative if and only if it has a nonnegative square root, corresponding to condition (d). Also, z is nonnegative if and only if it has a real square root, corresponding to condition (e). Finally, z is nonnegative if and only if there exists $w \in \mathbb{C}$ such that $z = \bar{w}w$, corresponding to condition (f). See Exercise 20 for another condition that is equivalent to being a positive operator.

*Because positive operators correspond to nonnegative numbers, better terminology would use the term nonnegative operators. However, operator theorists consistently call these positive operators, so we follow that custom. Some mathematicians use the term **positive semidefinite operator**, which means the same as positive operator.*

7.38 characterizations of positive operators

Let $T \in \mathcal{L}(V)$. Then the following are equivalent.

- (a) T is a positive operator.
- (b) T is self-adjoint and all eigenvalues of T are nonnegative.
- (c) With respect to some orthonormal basis of V , the matrix of T is a diagonal matrix with only nonnegative numbers on the diagonal.
- (d) T has a positive square root.
- (e) T has a self-adjoint square root.
- (f) $T = R^*R$ for some $R \in \mathcal{L}(V)$.

Proof We will prove that (a) $\Rightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (d) $\Rightarrow$ (e) $\Rightarrow$ (f) $\Rightarrow$ (a).

First suppose (a) holds, so that T is positive, which implies that T is self-adjoint (by definition of positive operator). To prove the other condition in (b), suppose λ is an eigenvalue of T . Let v be an eigenvector of T corresponding to λ . Then

$$0 \leq \langle Tv, v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle.$$

Thus λ is a nonnegative number. Hence (b) holds, showing that (a) implies (b).

Now suppose (b) holds, so that T is self-adjoint and all eigenvalues of T are nonnegative. By the spectral theorem (7.29 and 7.31), there is an orthonormal basis $e_1, \dots, e_n$ of V consisting of eigenvectors of T . Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of T corresponding to $e_1, \dots, e_n$; thus each λ_k is a nonnegative number. The matrix of T with respect to $e_1, \dots, e_n$ is the diagonal matrix with $\lambda_1, \dots, \lambda_n$ on the diagonal, which shows that (b) implies (c).

Now suppose (c) holds. Suppose $e_1, \dots, e_n$ is an orthonormal basis of V such that the matrix of T with respect to this basis is a diagonal matrix with nonnegative numbers $\lambda_1, \dots, \lambda_n$ on the diagonal. The linear map lemma (3.4) implies that there exists $R \in \mathcal{L}(V)$ such that

$$Re_k = \sqrt{\lambda_k} e_k$$

for each $k = 1, \dots, n$. As you should verify, R is a positive operator. Furthermore, $R^2 e_k = \lambda_k e_k = T e_k$ for each k , which implies that $R^2 = T$. Thus R is a positive square root of T . Hence (d) holds, which shows that (c) implies (d).

Every positive operator is self-adjoint (by definition of positive operator). Thus (d) implies (e).

Now suppose (e) holds, meaning that there exists a self-adjoint operator R on V such that $T = R^2$. Then $T = R^*R$ (because $R^* = R$). Hence (e) implies (f).

Finally, suppose (f) holds. Let $R \in \mathcal{L}(V)$ be such that $T = R^*R$. Then $T^* = (R^*R)^* = R^*(R^*)^* = R^*R = T$. Hence T is self-adjoint. To complete the proof that (a) holds, note that

$$\langle Tv, v \rangle = \langle R^*Rv, v \rangle = \langle Rv, Rv \rangle \geq 0$$

for every $v \in V$. Thus T is positive, showing that (f) implies (a). ■

Every nonnegative number has a unique nonnegative square root. The next result shows that positive operators enjoy a similar property.

7.39 each positive operator has only one positive square root

Every positive operator on V has a unique positive square root.

Proof Suppose $T \in \mathcal{L}(V)$ is positive. Suppose $v \in V$ is an eigenvector of T . Hence there exists a real number $\lambda \geq 0$ such that $Tv = \lambda v$.

Let R be a positive square root of T . We will prove that $Rv = \sqrt{\lambda}v$. This will imply that the behavior of R on the eigenvectors of T is uniquely determined. Because there is a basis of V consisting of eigenvectors of T (by the spectral theorem), this will imply that R is uniquely determined.

To prove that $Rv = \sqrt{\lambda}v$, note that the spectral theorem asserts that there is an orthonormal basis $e_1, \dots, e_n$ of V consisting of eigenvectors of R . Because R is a positive operator, all its eigenvalues are nonnegative. Thus there exist nonnegative numbers $\lambda_1, \dots, \lambda_n$ such that $Re_k = \sqrt{\lambda_k}e_k$ for each $k = 1, \dots, n$.

Because $e_1, \dots, e_n$ is a basis of V , we can write

$$v = a_1e_1 + \dots + a_n e_n$$

for some numbers $a_1, \dots, a_n \in \mathbb{F}$. Thus

$$Rv = a_1\sqrt{\lambda_1}e_1 + \dots + a_n\sqrt{\lambda_n}e_n.$$

Hence

$$\lambda v = Tv = R^2v = a_1\lambda_1e_1 + \dots + a_n\lambda_ne_n.$$

The equation above implies that

$$a_1\lambda e_1 + \dots + a_n\lambda e_n = a_1\lambda_1e_1 + \dots + a_n\lambda_ne_n.$$

Thus $a_k(\lambda - \lambda_k) = 0$ for each $k = 1, \dots, n$. Hence

$$v = \sum_{\{k: \lambda_k = \lambda\}} a_k e_k.$$

Thus

$$Rv = \sum_{\{k: \lambda_k = \lambda\}} a_k \sqrt{\lambda} e_k = \sqrt{\lambda}v,$$

as desired. ■

The notation defined below makes sense thanks to the result above.

7.40 notation: $\sqrt{T}$

For T a positive operator, $\sqrt{T}$ denotes the unique positive square root of T .

7.41 example: *square root of positive operators*

Define operators S, T on $\mathbf{R}^2$ (with the usual Euclidean inner product) by

$$S(x, y) = (x, 2y) \quad \text{and} \quad T(x, y) = (x + y, x + y).$$

Then with respect to the standard basis of $\mathbf{R}^2$ we have

$$7.42 \quad \mathcal{M}(S) = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \quad \text{and} \quad \mathcal{M}(T) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Each of these matrices equals its transpose; thus S and T are self-adjoint.

If $(x, y) \in \mathbf{R}^2$, then

$$\langle S(x, y), (x, y) \rangle = x^2 + 2y^2 \geq 0$$

and

$$\langle T(x, y), (x, y) \rangle = x^2 + 2xy + y^2 = (x + y)^2 \geq 0.$$

Thus S and T are positive operators.

The standard basis of $\mathbf{R}^2$ is an orthonormal basis consisting of eigenvectors of S . Note that

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$$

is an orthonormal basis of eigenvectors of T , with eigenvalue 2 for the first eigenvector and eigenvalue 0 for the second eigenvector. Thus $\sqrt{T}$ has the same eigenvectors, with eigenvalues $\sqrt{2}$ and 0.

You can verify that

$$\mathcal{M}(\sqrt{S}) = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{2} \end{pmatrix} \quad \text{and} \quad \mathcal{M}(\sqrt{T}) = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

with respect to the standard basis by showing that the squares of the matrices above are the matrices in 7.42 and that each matrix above is the matrix of a positive operator.

The statement of the next result does not involve a square root, but the clean proof makes nice use of the square root of a positive operator.

7.43 T positive and $\langle Tv, v \rangle = 0 \implies Tv = 0$

Suppose T is a positive operator on V and $v \in V$ is such that $\langle Tv, v \rangle = 0$. Then $Tv = 0$.

Proof We have

$$0 = \langle Tv, v \rangle = \langle \sqrt{T} \sqrt{T} v, v \rangle = \langle \sqrt{T} v, \sqrt{T} v \rangle = \|\sqrt{T} v\|^2.$$

Hence $\sqrt{T} v = 0$. Thus $Tv = \sqrt{T}(\sqrt{T} v) = 0$, as desired. ■

Exercises 7C

- 1 Suppose $T \in \mathcal{L}(V)$. Prove that if both T and $-T$ are positive operators, then $T = 0$.
- 2 Suppose $T \in \mathcal{L}(\mathbf{F}^4)$ is the operator whose matrix (with respect to the standard basis) is

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}.$$

Show that T is an invertible positive operator.

- 3 Suppose n is a positive integer and $T \in \mathcal{L}(\mathbf{F}^n)$ is the operator whose matrix (with respect to the standard basis) consists of all 1's. Show that T is a positive operator.
- 4 Suppose n is an integer with $n > 1$. Show that there exists an n -by- n matrix A such that all of the entries of A are positive numbers and $A = A^*$, but the operator on $\mathbf{F}^n$ whose matrix (with respect to the standard basis) equals A is not a positive operator.
- 5 Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Prove that T is a positive operator if and only if for every orthonormal basis $e_1, \dots, e_n$ of V , all entries on the diagonal of $\mathcal{M}(T, (e_1, \dots, e_n))$ are nonnegative numbers.
- 6 Prove that the sum of two positive operators on V is a positive operator.
- 7 Suppose $S \in \mathcal{L}(V)$ is an invertible positive operator and $T \in \mathcal{L}(V)$ is a positive operator. Prove that $S + T$ is invertible.
- 8 Suppose $T \in \mathcal{L}(V)$. Prove that T is a positive operator if and only if the pseudoinverse $T^\dagger$ is a positive operator.
- 9 Suppose $T \in \mathcal{L}(V)$ is a positive operator and $S \in \mathcal{L}(W, V)$. Prove that S^*TS is a positive operator on W .
- 10 Suppose T is a positive operator on V . Suppose $v, w \in V$ are such that

$$Tv = w \quad \text{and} \quad Tw = v.$$

Prove that $v = w$.

- 11 Suppose T is a positive operator on V and U is a subspace of V invariant under T . Prove that $T|_U \in \mathcal{L}(U)$ is a positive operator on U .
- 12 Suppose $T \in \mathcal{L}(V)$ is a positive operator. Prove that T^k is a positive operator for every positive integer k .

13 Suppose $T \in \mathcal{L}(V)$ is self-adjoint and $\alpha \in \mathbb{R}$.

- (a) Prove that $T - \alpha I$ is a positive operator if and only if α is less than or equal to every eigenvalue of T .
 (b) Prove that $\alpha I - T$ is a positive operator if and only if α is greater than or equal to every eigenvalue of T .

14 Suppose T is a positive operator on V and $v_1, \dots, v_m \in V$. Prove that

$$\sum_{j=1}^m \sum_{k=1}^m \langle T v_k, v_j \rangle \geq 0.$$

15 Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Prove that there exist positive operators $A, B \in \mathcal{L}(V)$ such that

$$T = A - B \quad \text{and} \quad \sqrt{T^*T} = A + B \quad \text{and} \quad AB = BA = 0.$$

16 Suppose T is a positive operator on V . Prove that

$$\text{null } \sqrt{T} = \text{null } T \quad \text{and} \quad \text{range } \sqrt{T} = \text{range } T.$$

17 Suppose that $T \in \mathcal{L}(V)$ is a positive operator. Prove that there exists a polynomial p with real coefficients such that $\sqrt{T} = p(T)$.

18 Suppose S and T are positive operators on V . Prove that ST is a positive operator if and only if S and T commute.

19 Show that the identity operator on $\mathbb{F}^2$ has infinitely many self-adjoint square roots.

20 Suppose $T \in \mathcal{L}(V)$ and $e_1, \dots, e_n$ is an orthonormal basis of V . Prove that T is a positive operator if and only if there exist $v_1, \dots, v_n \in V$ such that

$$\langle T e_k, e_j \rangle = \langle v_k, v_j \rangle$$

for all $j, k = 1, \dots, n$.

The numbers $\{\langle T e_k, e_j \rangle\}_{j,k=1,\dots,n}$ are the entries in the matrix of T with respect to the orthonormal basis $e_1, \dots, e_n$.

21 Suppose n is a positive integer. The n -by- n Hilbert matrix is the n -by- n matrix whose entry in row j , column k is $\frac{1}{j+k-1}$. Suppose $T \in \mathcal{L}(V)$ is an operator whose matrix with respect to some orthonormal basis of V is the n -by- n Hilbert matrix. Prove that T is a positive invertible operator.

Example: The 4-by-4 Hilbert matrix is

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \frac{1}{7} \end{pmatrix}.$$

- 22** Suppose $T \in \mathcal{L}(V)$ is a positive operator and $u \in V$ is such that $\|u\| = 1$ and $\|Tu\| \geq \|Tv\|$ for all $v \in V$ with $\|v\| = 1$. Show that u is an eigenvector of T corresponding to the largest eigenvalue of T .
- 23** For $T \in \mathcal{L}(V)$ and $u, v \in V$, define $\langle u, v \rangle_T$ by $\langle u, v \rangle_T = \langle Tu, v \rangle$.
- (a) Suppose $T \in \mathcal{L}(V)$. Prove that $\langle \cdot, \cdot \rangle_T$ is an inner product on V if and only if T is an invertible positive operator (with respect to the original inner product $\langle \cdot, \cdot \rangle$).
- (b) Prove that every inner product on V is of the form $\langle \cdot, \cdot \rangle_T$ for some positive invertible operator $T \in \mathcal{L}(V)$.
- 24** Suppose S and T are positive operators on V . Prove that
- $$\text{null}(S + T) = \text{null } S \cap \text{null } T.$$
- 25** Let T be the second derivative operator in Exercise 31(b) in Section 7A. Show that $-T$ is a positive operator.